Yoel Kim

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Research Interests

- Software and its engineering → Formal software verification; Model checking; Embedded software;
- Theory of Computation → Automata over infinite objects; Abstraction.

Education

Ph.D. in Computer Science and Engineering, Kyungpook National University — *Mar 2023 – Feb 2027 (expected)*

- Advisor: Yunja Choi
- Thesis: Data-Assisted Abstraction Techniques for Embedded Software Verification

M.S. in Computer Science and Engineering, Kyungpook National University — *Mar 2021 – Feb 2023*

- Advisor: Yunja Choi
- Thesis: An Automated Stub Generation Approach Using Program Synthesis for Software Verification

B.S. in Computer Science and Engineering, Kyungpook National University — *Mar 2017 – Feb 2021*

International Conferences

1. Active Learning of Symbolic Automata for Reactive Programs via Dynamic Symbolic Mapper
   Yoel Kim and Yunja Choi
   FSE'26: *ACM International Conference on the Foundations of Software Engineering*

2. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software
   Yoel Kim and Yunja Choi
   FSE'24: *ACM International Conference on the Foundations of Software Engineering*

Domestic Conferences

1. LLM-Based State Machine Generation Technique for Reactive Systems and Its Performance Evaluation
   Seungbin Choi, Yoel Kim, and Yunja Choi
   KCSE'26: *Korea Conference on Software Engineering* * 우수 단편 논문

2. An Approach of Incremental Constraint Extraction Based on I/O Examples for Automatic Stub Generation
   Yoel Kim and Yunja Choi
   KCSE'23: *Korea Conference on Software Engineering* * 최우수 단편 논문

3. A Case Study to Improve the Efficiency of Model Checking in Embedded Software Using Program Synthesis
   Yoel Kim and Yunja Choi
   KSC'21: *Korea Software Congress*

4. A Case Study on the Performance of Program Synthesis in Embedded Software Domain
   Yoel Kim and Yunja Choi
   KCSE'21: *Korea Conference on Software Engineering*

Invited Talks

1. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference session. KCC'25. *Jeju, Korea. Jul 4, 2025.*

2. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Top Conference session. KCSE'25. *Pyeongchang, Korea. Jan 22, 2025.*

3. PBE-Based Selective Abstraction and Refinement for Efficient Property Falsification of Embedded Software. Research paper presentation. FSE'24. *Porto de Galinhas, Ipojuca, Pernambuco, Brazil. Jul 17, 2024.*

Experiences

Teaching Assistant (at Kyungpook National University):

- ITEC0414: Software Testing Theory (Spring 2022, 2023, 2024).
- COMP0224: Software Design (Fall 2021, 2022).
- COMP0216: Data Structure Applications (Spring 2021).